

WTW 158 Worksheet 9 on L'Hospital's rule, curve sketching and antiderivatives

A. TEXTBOOK PROBLEMS

p 311 no 9, 15, 17, 25, 31, 53, 55, 57, 59

p 300 no 25, 29

p 321 no 7

p 355 no 1- 21 (odd numbers) 33, 39, 49

B. MORE PROBLEMS

For the function $f(x) = x - 2 \sin x$, $0 \leq x \leq 2\pi$, find:

(1) the intervals on which f is increasing / decreasing

(2) extremes

(3) the intervals on which f is concave up / concave down

(4) possible point(s) of inflection

C. TEST QUESTIONS

1. Find the interval(s) on which the function $f(x) = xe^{-x^2}$ is increasing.

2. $\lim_{x \rightarrow 0} (1 + kx)^{\frac{1}{x}} =$

[a] 1	[b] 0	[c] e^k	[d] e^{-k}	[e] doesn't exist
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D. SELECTED ANSWERS

A. p 311 9. 6 15. 2 17. $\frac{1}{4}$ 25. 3 31. 1 53 $\frac{1}{2}$ 55. ∞ 57. 1 59. e^{-2}

p 300 25 & 29 See textbook pA82

p 321 7. See textbook pA84

p 355 1 - 21 (Odd) See textbook p A90

B.(1) Decreasing on $(0, \frac{\pi}{3})$ and on $(\frac{5\pi}{3}, 2\pi)$, increasing on $(\frac{\pi}{3}, \frac{5\pi}{3})$

(2) $f(\frac{\pi}{3}) = \frac{\pi}{3} - \sqrt{3}$ is a local minimum, $f(\frac{5\pi}{3}) = \frac{5\pi}{3} + \sqrt{3}$ is a local / abs maximum

(3) f is concave up on $(0, \pi)$, f is concave down on $(\pi, 2\pi)$

(4) Inflection point (π, π)

C. 1. $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ 2. [c]

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